

Distributed Lupus in Fabula

A Replicated Real-Time Multiplayer Game Platform

Final Report for the Distributed Systems Course — A.Y. 2025/2026

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Distributed Lupus in Fabula is a real-time multiplayer web application that brings the classic social deduction game Lupus in Fabula online, allowing groups of players to create or join a game lobby from any browser and play a full match together over the internet. The system is composed of a Vue 3 single-page frontend and a FastAPI/Socket.IO backend, communicating via REST for stateless operations and via WebSocket for real-time, low-latency interactions such as votes, phase transitions, and private role actions. Game and lobby state is kept in Redis rather than a traditional database, acting both as a shared source of truth across backend replicas and as a Pub/Sub broker that lets events generated on one instance reach clients connected to any other instance. This design allows the backend to run as multiple concurrent, interchangeable replicas behind a reverse proxy, supporting many simultaneous matches while keeping every player's view of the game synchronized in real time.

Disclaimer

During the preparation of this work, the authors used LLMs to refine syntax and correct linguistic errors. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the final report/artifact.

1 Concept

1.1 Type of Product

The project is a web application, specifically a real-time multiplayer game inspired by the traditional social deduction game *Lupus in Fabula*. It is a full-stack distributed system composed of:

- a **frontend**: a Vue 3 Single Page Application (SPA), accessible via browser, with no installation required;
- a **backend**: a set of stateless, horizontally replicated web services built with FastAPI and Socket.IO, responsible for game logic, session management, and real-time event propagation;
- a **shared data layer**: Redis, used both as a low-latency state store and as a Pub/Sub message broker connecting the backend replicas.

These are complemented by an infrastructure layer (reverse proxy / ingress, monitoring stack) that is part of the deliverable (section 3.2). The whole system is designed to be deployed as a multi-container distributed application rather than as a single monolithic server.

1.2 Use Case Description

What the software does. The application lets a group of 5 or more players create or join a virtual “lobby”, wait until everyone is ready, and then play a phase-based game of hidden roles and social deduction. Players are secretly split into two factions: *wolves*, who eliminate one villager per night, and *villagers* (possibly including a *seer*, who can privately inspect one player’s faction each night), who try to identify and lynch the wolves through a public vote. Once a match starts, the game state moves automatically through a fixed sequence of phases driven by server-side timers:

LOBBY → NIGHT → DAY → VOTING → ... → ENDED.

During NIGHT, players with special roles perform hidden actions that the server resolves before the next day begins; during DAY/VOTING, players discuss and vote to eliminate a suspect. The match ends automatically once a win condition is met, and the server is authoritative throughout: it assigns hidden roles, collects votes and night actions, computes eliminations, and detects the winner.

Where the users are. Players are geographically dispersed (or at least not necessarily co-located): each player connects independently from their own browser, over the public Internet, to a shared game session identified by a room/lobby code. The service itself runs in a datacenter (a container orchestration cluster).

When and how frequently they interact. Interaction is continuous and real-time for the whole duration of a match (several minutes to tens of minutes), not just at fixed request/response intervals. Players must react within phase time limits (voting windows, night-action windows), so the system needs low-latency, event-driven communication rather than periodic polling.

How they interact. All interaction happens through a web browser via the Vue SPA. The frontend communicates with the backend in two complementary ways:

- **REST** calls for stateless operations such as creating a lobby, joining a lobby, or fetching the current game state;
- **WebSocket** (Socket.IO) for real-time, bidirectional events: votes, phase changes, chat, private role actions, disconnect/reconnect notifications.

Does the system store user data? Which, and where? The system does not use a persistent relational database for user accounts, since there is no long-term profile storage. Instead it relies on Redis as a shared, TTL-based store for all session-scoped game data:

- room/lobby state and configuration;
- the list of connected players and their ready status;
- assigned roles (Wolf, Seer, Villager);
- votes cast during the DAY/VOTING phase and private night actions;
- phase timers and current phase;
- host identity.

Clients are never trusted with authoritative data (a client that knew the role assignments could cheat), and state expires automatically (TTL) hours after the last update, so no long-term personal data is persisted. On the client side, the browser retains a minimal amount of session data in local storage, purely to support reconnection and continuity across page reloads or brief disconnects.

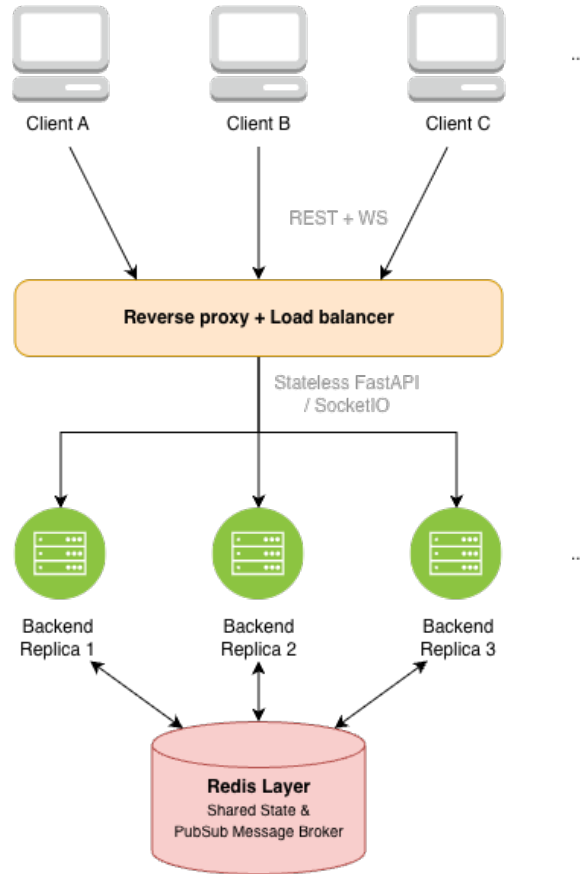


Figure 1: High-level architecture: clients connect through a reverse proxy to stateless backend replicas, which share game state and propagate events through Redis.

Roles. The system distinguishes several actor roles at different levels:

- *Application-level roles:* **Villager**, **Wolf**, **Seer**, each with different permitted actions and visibility (e.g. wolves share a private night chat and see each other's votes; the seer can inspect one player per night);
- *Session-level roles:* the **host**, who configures the match, can remove players from the lobby, starts the game, and restarts it once ended, versus **regular players**, who can only join, ready up, vote, and act according to their assigned role; the host role migrates automatically if the current host disconnects;
- *System role:* the backend itself acts as an autonomous **orchestrator**. It owns the authoritative game state, advances phases via timers, resolves votes and night actions, and enforces which actions are legal at any given moment.

1.3 Why Distribution Is Needed

Distribution in this project is not motivated by geographic proximity to users, but primarily by scalability, real-time synchronization, and fault tolerance:

- **Horizontal scalability / resource sharing.** The backend is designed to run as multiple stateless replicas behind a reverse proxy, so that many concurrent lobbies and matches can be served in parallel without a single process becoming a bottleneck. Because game state lives in Redis rather than in each process's memory, any replica can handle requests for any room, allowing the system to scale out simply by adding more backend instances.
- **Shared state across replicas.** Since each backend instance is stateless with respect to game data, Redis acts as the single source of truth for lobby/game state. This is what makes horizontal scaling actually work: a player's WebSocket connection can be handled by one replica while another replica processes a REST call for the same match, and both see a consistent view of the state.
- **Cross-replica real-time propagation.** Because clients belonging to the same room can be connected to different backend replicas, a simple in-process event emitter is not sufficient. The system uses Redis Pub/Sub to broadcast game events (vote updates, phase transitions, chat messages, etc.) across all replicas, ensuring that every client in a room receives updates regardless of which instance generated them or which instance is handling their socket.
- **Fault tolerance and continuity.** The backend includes explicit mechanisms for reconnecting to Redis, retrying failed operations, sweeping/recovering timers, and reconciling state after temporary inconsistencies (anti-entropy). This matters because a real-time game cannot simply crash or stall if a single backend replica fails, a Redis connection drops momentarily, or a player's browser disconnects and reconnects mid-match. The match state and timers must survive these events.
- **Reduced dependency on a single instance.** Because the game state is decoupled from any specific process, a backend replica can be restarted, replaced, or scaled down/up without necessarily terminating ongoing matches, and a reconnecting player can resume their session against a different replica than the one they were originally connected to.

In summary, this system is distributed primarily to support many concurrent matches, keep real-time state synchronized across independent backend replicas, and remain resilient to instance failures and client disconnections.

2 Requirements Elicitation and Analysis

This section specifies *what* the system must do, independently of the implementation strategies later adopted. Requirements are grouped into functional, non-functional, and

implementation requirements, each with a unique identifier and an associated acceptance criterion (AC) used during validation.

2.1 Glossary

Room (or lobby) An independent game session, identified by a user-chosen code, hosting one match at a time.

Host The player who administers a room: configures the match, starts it, kicks players, and restarts the game once it has ended.

Phase A stage of the game loop: LOBBY, NIGHT, DAY, VOTING, ENDED.

Role The secret game faction assigned to a player: *wolf*, *seer*, or *villager*.

Replica (or instance) One of the interchangeable backend server processes serving client connections.

Snapshot The complete, authoritative representation of a room's state, as stored server-side and sent to (re)connecting clients.

2.2 Functional Requirements

FR1 — Room creation and joining. A user can create or join a room by choosing a nickname and a room code; the first user entering a room becomes its host. *AC: two users entering the same code end up in the same lobby; exactly one host exists, even if two users create the same room simultaneously.*

FR2 — Lobby browsing. A user can list the currently open rooms (those still in LOBBY phase) with their host and player count. *AC: a room in progress or ended does not appear in the list; an open room appears regardless of which replica serves the request.*

FR3 — Match configuration. The host, and only the host, can configure the number of wolves and seers while in LOBBY. Configurations that break the game balance are rejected (at least 5 players, at least 1 wolf, at least 1 plain villager left). *AC: a non-host attempt or an invalid configuration is refused with an error message; a valid change is visible to all lobby members.*

FR4 — Readiness and game start. Players declare themselves ready; the host can start the match only when all connected players are ready and the configuration is valid. *AC: a start request with a non-ready player is rejected; on start, every player receives their secret role.*

FR5 — Secret role assignment. At game start the system randomly assigns roles according to the configuration; each player learns *only* their own role (wolves also learn their teammates). *AC: no message delivered to a villager's client ever contains another player's role before game end.*

- FR6 — Game loop and timers.** The game alternates NIGHT → DAY → VOTING phases, each bounded by a server-side timer; when the timer expires the phase advances automatically for everyone. *AC: phases advance without any client input; all clients converge on the same phase and deadline.*
- FR7 — Public vote.** During VOTING every living player can vote a living target (or skip); votes are public, ties or skip-majority cause no elimination; the most-voted player is eliminated and their role revealed. *AC: dead or disconnected players cannot vote; the tally matches the recorded votes even when voters are connected to different replicas.*
- FR8 — Night actions.** During NIGHT, wolves privately agree on a victim (seeing each other's votes), and the seer privately inspects one player's faction. *AC: night choices are never visible to other players; the seer's result is delivered only to the seer, even if connected to a different replica than the one computing it.*
- FR9 — Win detection.** The system detects the end of the game as soon as a faction wins (no wolves left, or wolves $\geq$ non-wolves) — including when the decisive event is a player abandoning the match — and reveals all roles. *AC: the ended screen shows winner and roles to every player; no further game action is accepted.*
- FR10 — Chat.** Players in a room can exchange real-time chat messages over restricted-visibility channels (a global channel during the day, a wolves-only channel at night, and a dead-players channel); system announcements (e.g. a player fleeing mid-game) appear in the same feed. *AC: a message on a restricted channel is rendered only by the clients entitled to see it, and a message sent by a player on replica A reaches entitled players on replica B.*
- FR11 — Reconnection.** A player who loses the connection mid-game can rejoin the same room with the same nickname and receive the current authoritative state (including their role). *AC: after a page reload during a match, the player sees the current phase, timer, and their own role, even when reconnected to a different replica.*
- FR12 — Host migration.** If the host disconnects, another connected player is automatically promoted host. *AC: the lobby never remains without a host while at least one player is connected.*
- FR13 — Room administration.** The host can kick a player from the lobby, and can restart a finished game bringing everyone back to the lobby (with roles, votes, and readiness reset, and disconnected players cleaned up). *AC: a kicked player is removed from every registry and notified; after restart, no “ghost” player remains and nobody is marked ready.*
- FR14 — Access control to running games.** A user who is not part of a match cannot join a room where a game is in progress or ended, and a nickname already connected

to a room cannot be reused concurrently. *AC: the connection is refused with an explanatory message.*

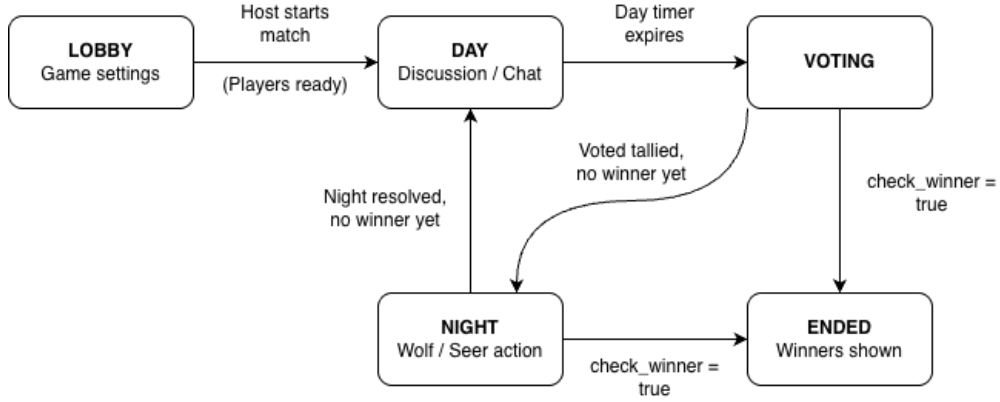


Figure 2: State machine of the match phases. Transitions are driven automatically by server-side timers and by `check_winner()`.

2.3 Non-Functional Requirements

NFR1 — Replication transparency. Clients must not know, or care, which backend replica serves them: any replica must be able to serve any room, and two players in the same room must be able to play together while connected to different replicas. *AC: the multi-instance test suite passes with 2+ replicas behind the load balancer.*

NFR2 — Fault tolerance. The crash of a single backend replica must not destroy any running match: affected clients reconnect and resume within seconds; matches served by other replicas must not be affected. Phase progression must not depend on the survival of the replica that scheduled the current phase timer. *AC: killing one replica during a match, the game reaches its natural end.*

NFR3 — Convergence (eventual consistency). All clients of a room must converge to the same authoritative game state: temporarily missed events must be repaired automatically within seconds, without user intervention. *AC: a client that missed a broadcast converges to the authoritative snapshot within one anti-entropy period (~ 10 s).*

NFR4 — Safety of game rules. Game-rule invariants must hold despite concurrency and replication: no lost updates on shared state, at most one host per room, at most one phase advancement per deadline (no double eliminations). *AC: the distributed-safety test suite passes with concurrent clients on multiple replicas.*

NFR5 — Scalability. Capacity (concurrent connections/rooms) must grow by adding backend replicas, with no code changes and no single-replica bottleneck in the

service tier; the deployment must support automatic horizontal scaling. *AC: under load, the autoscaler increases replicas and new connections are spread across them.*

NFR6 — Soft real-time interactivity. Game events must reach all players of a room with sub-second perceived latency under nominal load. *AC: load tests report end-to-end event fan-out latencies well below 1 s at the target load.*

NFR7 — Observability. Operators must be able to inspect the health and behaviour of every component (connections per replica, Pub/Sub traffic, error rates, resource usage) through metrics, dashboards, and alerts. *AC: Grafana dashboards display per-replica WebSocket and Redis metrics; alert rules fire on component failure.*

NFR8 — Zero-downtime updates. Deploying a new backend version must not interrupt the service. *AC: a rolling update completes with no failed health checks and no window with zero ready replicas.*

NFR9 — Usability. The interface must let a non-technical user join and play a match without understanding the system’s internal details. *AC: a user can complete lobby creation/joining, the lobby phase, and the match using only a browser and a room code.*

2.4 Implementation Requirements

By design (section 3) the system needs an asynchronous, WebSocket-capable service runtime, a shared low-latency store with publish/subscribe capabilities, and reproducible containerised deployment. The concrete constraints below descend from those needs plus course/administrative context:

IR1 The system shall be containerised and runnable both with Docker Compose (local, single machine) and on Kubernetes (cluster), since deployment automation and orchestration are learning goals of the project. *AC: both **docker compose up** and the provided Kubernetes manifests bring up the full platform.*

IR2 The backend shall expose Prometheus-compatible metrics, as Prometheus + Grafana is the de-facto standard open-source monitoring stack and fits the zero-budget constraint. *AC: **/metrics** is scraped successfully by the provided Prometheus configuration and visualised in Grafana.*

IR3 Only free / open-source components shall be used (no managed cloud services), for economic reasons and local reproducibility on student machines. *AC: the whole platform runs on a laptop via minikube.*

Programming languages and frameworks (Python/FastAPI, Vue.js, Redis) were *not* imposed a priori; they emerge as consequences of the design and are motivated in section 4.

2.5 Relevant Distributed System Features

We assess which classic distributed-system concerns [10] are relevant for this project, and to what extent.

Transparency — central. *Replication* and *access* transparency are the core of the project: users must perceive a single logical game server, while connections are actually served by interchangeable replicas (NFR1). *Failure* transparency is pursued for server-side faults (a replica crash appears as a brief reconnection, not a lost match — NFR2). *Location* transparency is provided by naming services and virtual hosts (clients address `game.local`, never a specific pod). Migration, relocation, and persistence transparency are not relevant: components do not move at runtime and no long-term persistence is exposed to users.

Fault tolerance / dependability — central. The service tier must tolerate crash failures (fail-stop) of any single replica, and temporary unavailability of the message bus. Uninterrupted service is required at match granularity: losing a running match is the failure to avoid. Recovery must be fast (seconds), since matches are paced by timers. Byzantine behaviour of *servers* is out of scope (all replicas are trusted, deployed from the same image); malicious *clients* are considered only at the game-rule level (the server is authoritative, FR5/FR14). Durability beyond a match is explicitly *not* required: game state is ephemeral, which licenses an in-memory store with TTL-based garbage collection rather than a durable database.

Scalability — central. The system must scale *out* with the number of concurrent players and rooms (NFR5). Rooms are independent, so the workload partitions naturally; the stateless service tier scales horizontally while the shared store stays small (per-room state is a few KB) and is not the bottleneck at the target scale. Geographic (multi-datacenter) scalability is out of scope.

Security / trust — limited relevance. No sensitive or personal data is stored (nicknames only), so confidentiality and GDPR-style concerns are minimal. What matters is *authorization* at the application level: host-only administrative actions (FR3, FR13) and role-based information hiding during a match (FR5, FR8) — trust is placed in the server, never in clients. Strong authentication is deliberately absent (casual, friction-free party game); the perimeter is protected by rate limiting and security headers at the proxy. TLS is prepared for but not enabled in the local deployments.

Resource sharing — central. All replicas share the authoritative state store and the inter-replica event bus; all rooms share the replica pool. This is precisely what requires the coordination and synchronisation mechanisms (atomic updates, distributed locks) described in section 3.6.

Performance / concurrency — central. The workload is massively concurrent (many rooms, many sockets) but computationally light: the challenge is efficient I/O

multiplexing and fan-out, not CPU. This drives the choice of an asynchronous, event-loop-based runtime and of connection-aware load balancing; per-message bandwidth is negligible (small JSON events).

Conversely, some concerns are deliberately *non-central*: geographical distribution (no geo-aware routing or replication is required), heterogeneity beyond the standard web stack (all components speak HTTP/WebSocket/JSON, with no integration of complex third-party systems), and a strong security model (no critical personal data or financial transactions are handled).

3 Design

This chapter explains the strategies used to meet the requirements identified in the analysis. The design is described in technology-neutral terms wherever possible; concrete technology bindings are deferred to section 4.

3.1 Architecture

The system combines two classic architectural styles over a client/server organisation.

Layered (three-tier) backbone. At the coarse grain the system is a *three-tier layered* architecture: a **presentation tier** (the SPA running in each player’s browser), an **application tier** (a pool of identical, *stateless* game-server replicas behind a load-balancing proxy), and a **data tier** (a shared in-memory store holding all authoritative state). The backend does not treat its in-memory representation as authoritative; it reconstructs domain state from Redis on demand and uses Pub/Sub purely as a notification mechanism, never as a source of truth. Layering gives separation of concerns and lets each tier scale independently: the application tier scales horizontally precisely *because* it is stateless — “stateless servers + stateful store” is the standard recipe for service redundancy [3], and it directly serves NFR1/NFR5.

Event-based core. Within a match interaction is not request/response: the dominant flow is *events* (a vote was cast, a phase changed, a player was eliminated) that must be *pushed* to all interested parties with low latency (NFR6). The architecture is therefore *event-based* at two levels:

- between server and clients, each room acts as a topic: clients subscribe by joining the room, and the server pushes domain events to all of its members over persistent connections;
- between server replicas, a *publish/subscribe event bus* (one logical channel per room, hosted by the data tier) propagates every event to the other replicas, which relay it to their locally connected clients.

The publish/subscribe connector is what makes replication transparent (NFR1): replicas are *referentially and spatially uncoupled* — they never address each other, do not know how many peers exist, and need no membership protocol; the set of replicas can change at any time (autoscaler, crash, rolling update) without reconfiguration [10].

Client-side layering. The frontend is itself a Single-Page Application organised in three internal layers, so that the client stays event-driven and *stateless with respect to game logic*: **views** present state and collect input but never touch the network; **stores** hold the reactive client state, register listeners for server events, and expose the actions that emit commands; a single **communication layer** (a Socket.IO wrapper) is the only component that touches the network. Because the client holds no authoritative state, it can always be re-synchronised from the server after a fault and behaves identically regardless of the instance it is attached to (N2/N3 of the client scope).

Why not alternatives. A single stateful game server (state in process memory, no bus) would be simpler but violates NFR2/NFR5: a single point of failure and a vertical-scaling dead end. Sticky sessions (pinning a whole room to one replica) would remove the need for the bus but reintroduce the same problem at room granularity — a replica crash would still destroy its rooms — and defeat connection-level load balancing. Peer-to-peer distribution among clients was discarded because hidden roles (FR5) demand an authoritative, trusted party that clients cannot inspect. Finally, consensus-based state-machine replication across replicas [8] was deliberately avoided: it would provide stronger consistency than the game needs, at the price of latency and complexity (section 3.6 argues that a shared atomic store plus eventual client convergence is sufficient).

3.2 Infrastructure

Components. The deployed system comprises the following components:

Clients (N). One browser per player, holding a single persistent bidirectional connection plus occasional REST calls.

Reverse proxy / load balancer (1). The single entry point. It routes by path — real-time traffic and the REST API to the backend pool, everything else to the frontend — terminates and upgrades WebSocket connections, applies per-IP rate limiting (section 3.9), and balances *new connections* across backend replicas round-robin. In the orchestrated deployment this role is played by the cluster’s ingress controller.

Frontend static server (1–2). Serves the SPA assets. Trivially replicable since it holds no state; at page load it injects the runtime configuration (the backend origin) so the same image works in every environment.

Backend replicas (2–8). Identical stateless game servers. Two is the floor (no single point of failure in the service tier, NFR2); the ceiling is set by the autoscaler (NFR5). Each replica has a unique *instance identifier* used for event deduplication, distributed-lock ownership, and per-replica metrics.

Shared store / message broker (1). A single in-memory data store playing *two* roles at once: the authoritative state database (key–value snapshots, hashes for player registries and vote maps) and the inter-replica publish/subscribe broker (one channel per room plus a global channel). Collapsing the two roles keeps the infrastructure minimal (IR3). The store is a deliberate single stateful instance — section 3.7 discusses how its temporary unavailability is tolerated and section 9 sketches its high-availability evolution.

Monitoring stack (3). A metrics scraper/time-series database (pull-based collection from every backend replica and from a store-metrics exporter), a dashboard server, and the exporter itself (NFR7, IR2). Pull-based scraping was preferred to push because replicas come and go (autoscaling): the scraper discovers targets dynamically instead of each ephemeral replica registering itself.

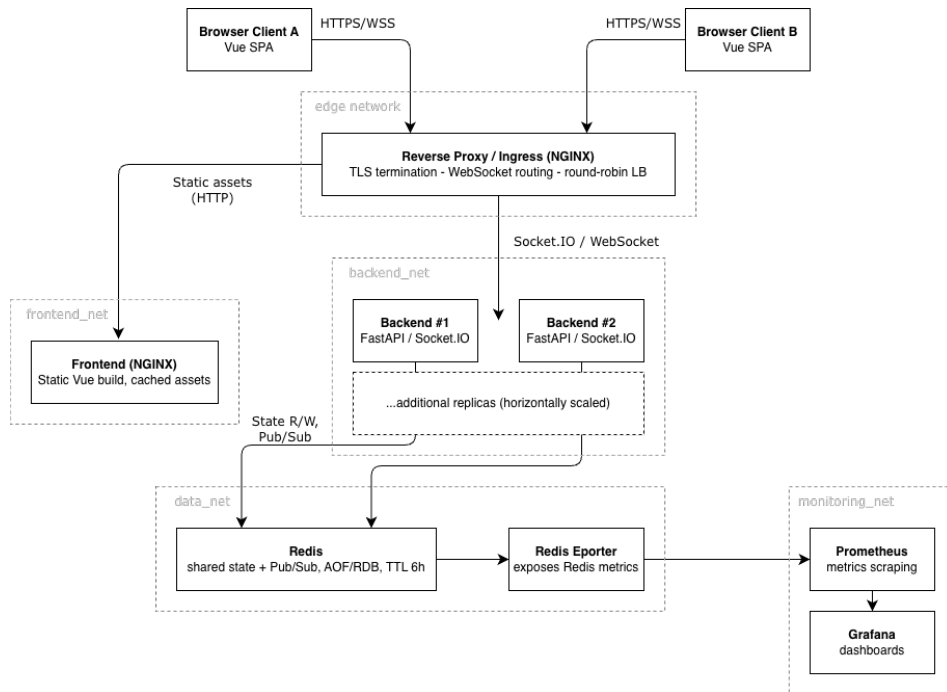


Figure 3: Component diagram: logical components and their dependencies — browser clients, the reverse proxy / ingress, the frontend static server, the pool of backend replicas, the shared Redis store + broker, and the monitoring stack.

Network placement. All server-side components run in the same cluster/datacenter — distribution here serves fault tolerance and scalability, not geo-proximity (section 2.5). Latency-critical hops (replica $\leftrightarrow$ store) are intra-cluster and sub-millisecond. Clients reach the cluster over the public Internet through the single proxy entry point. Internally the network is segmented into four bridge networks (table 1): the proxy can reach the frontend and the backend pool but *not* the store; the store sits on two networks,

reachable by the backends (data path) and by the observability stack (monitoring path). The edge tier and the store share no network at all, which confines the blast radius of a compromised edge component. In Kubernetes the same segmentation is reproduced through separate Services and namespaced routing, with backend and Redis exposed only as internal `ClusterIP` services and external traffic entering exclusively through the NGINX Ingress.

Table 1: Network membership of each component in the Docker Compose topology. Two components can talk at the network layer *iff* they share a bridge network. The edge proxy shares no network with the store, which confines the blast radius of a compromised edge; the store is reachable on `data_net` by the backends and on `monitoring_net` by the observability stack.

Component	<i>Bridge network</i>			
	frontend	backend	data	monitoring
Reverse proxy / Ingress	•	•		
Frontend static server	•			
Backend replica ($\times N$)		•	•	
Redis (store + bus)			•	•
Prometheus		•		•
Grafana				•
redis-exporter				•

Naming and discovery. Components find each other by *name*, never by address: clients use one virtual host (e.g. `game.local`) for everything — SPA, API, and real-time endpoint differ only by path, so the SPA needs no per-environment endpoint configuration (location transparency); the proxy resolves the backend pool through the platform’s internal DNS *at request time*, so replicas added or removed by scaling are picked up without reloading; backends reach the store through a stable service name configured via environment (twelve-factor style); and replicas do *not* discover each other at all — the publish/subscribe bus makes peer discovery unnecessary. The frontend receives the backend’s reachable address at runtime through an environment variable (`WS_URL`), so a single frontend image is deployed against different backend endpoints.

3.3 Modelling

Domain entities. The domain model is small and centred on the room:

Room an independent match container, identified by its code; it owns all other entities and is the unit of isolation (and of publish/subscribe topicality).

Player nickname, secret *role*, liveness (*alive*), connectivity (*connected*), readiness, host flag, and per-round action flags (*has voted*, *has acted*).

GameState the room’s phase (LOBBY, NIGHT, DAY, VOTING, ENDED), round number, phase deadline (*timer end*), match configuration (wolf/seer counts), host identity, ready set, winner, and a monotonically increasing *state version*.

Votes two separate ballots: the public day vote (voter → target, including a distinguished *skip* target) and the secret wolf vote; plus the seer’s nightly inspection choice.

Chat message sender, text, and channel; chat is transient — relayed, never stored.

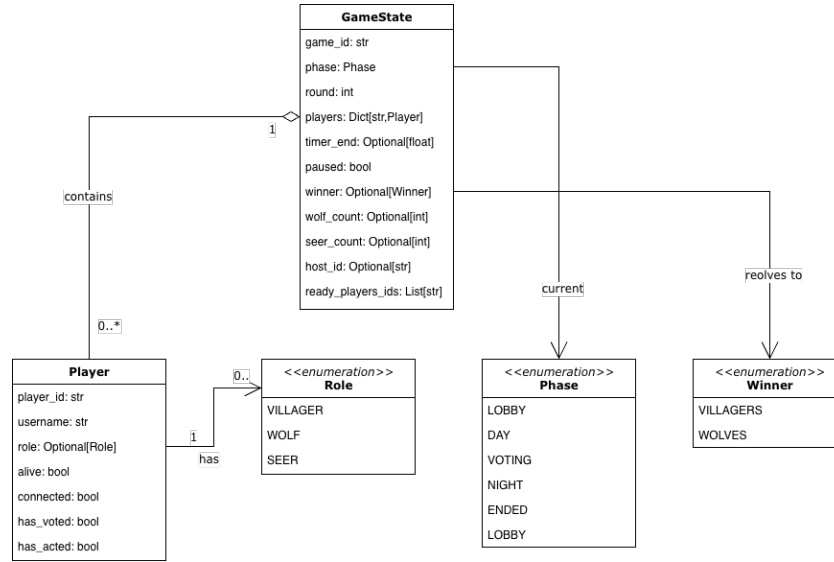


Figure 4: Domain class diagram: the room owns players, game state, ballots, and the phase timer; roles and phases are enumerations.

Mapping entities to infrastructure. All authoritative entity state lives in the shared store, namespaced per room and scoped under a room-level key (no query ever spans multiple rooms except the lobby list); nothing authoritative lives in a replica’s memory. Clients hold only a *view*: a local copy of the last received snapshot/events used for rendering, partitioned into three reactive stores — a *lobby store* (room code, players, host, ready set, role setup), a *game store* (phase machine, own role, phase timer, vote tally, pause state, outcome), and a *chat store* (messages, active channel, seen-message-id set for de-duplication). Replicas hold only *ephemeral operational* state — the registry of locally connected sockets and the in-memory phase-timer tasks — reconstructible and tolerated to be lost (section 3.7). Chat messages are the one entity deliberately *not* stored anywhere. All stored state carries a time-to-live measured in hours, so abandoned rooms garbage-collect themselves.

Domain events and message sorts. The system is event-sourced at the *interaction* level (not at storage level): every relevant fact is a named event — room membership

(player joined, left, host changed, kicked), lobby (settings updated, ready changed, game start), game progression (role assigned [private], phase changed, vote update, player eliminated/killed, no elimination, seer result [private], game ended), synchronisation (full authoritative snapshot), and system (chat message, error). Three sorts of messages are exchanged: **Commands** (client → server: cast vote, wolf vote, seer action, update settings, set ready, start, kick, restart) are validated against authoritative state and can be rejected (an *error* event answers the issuer only); **Events** (server → clients, and server → server via the bus) notify state changes — events between replicas carry an envelope with a unique id, room id, *sender replica id* (for deduplication), timestamp, and optionally a *target client id* marking a private unicast; **Queries** (lobby list, health) never modify state and can be answered by any replica. Separating commands (intents, which may fail) from events (committed facts) keeps the authorization logic of section 3.9 straightforward.

3.4 Interaction

Interaction combines four patterns, each used where its coupling fits.

Request/response is used only for queries (lobby list, health checks) and for serving static assets, plus a single REST call to fetch the initial game state when entering the game screen. Any replica can play the responder role.

Publish/subscribe, twice. The core gameplay flow enacts publish/subscribe at two nested levels:

1. *client-facing*: joining room *R* subscribes a client to *R*'s event stream over its persistent connection; the serving replica is the broker for its local subscribers;
2. *replica-facing*: each replica subscribes to the bus channel of every room for which it currently hosts at least one client (unsubscribing when its last local member leaves). Every event a replica produces is both emitted to its local subscribers *and* published on the room channel.

A replica receiving a bus message relays it to its local room members — *unless* the message's *sender id* is its own, in which case it drops it (sender-based **deduplication**), preventing the double delivery that would otherwise occur on the originating replica. Local emission is done before/besides publication on purpose: clients on the originating replica get minimal latency, and a bus outage degrades cross-replica delivery only (section 3.7). An important client-side rule complements this: the client *never assumes its command succeeded* until it receives the corresponding confirmation event — casting a vote does not update the local tally directly; the tally updates only when the server broadcasts the `vote_update`. Table 2 summarises the event contract.

Scope	Received (server → client)		Emitted (client → server)
Lobby	game_state_sync, player_left, role_setup_updated, kicked	player_joined, player_ready, room_closed,	lobby:player_ready, lobby:update_settings, lobby:start_game, kick_player
Game	game_start / phase_changed, vote_update, player_killed, no_elimination, game_paused, game_resumed	start_game, role_assigned, player_eliminated, seer_result, game_ended,	cast_vote, wolf_vote, seer_action, return_to_lobby
Chat	chat_message		chat_message

Table 2: Socket.IO event contract between client and server (exact names shared with the backend).

The typical fan-out of a game action (a vote) is: (1) client C_1 on replica A sends the *cast vote* command; (2) A validates it against the store, records the vote atomically, and builds the *vote update* event; (3) A emits it to its local room members and publishes it on the room channel; (4) replica B receives it from the bus, sees a foreign sender id, and relays it to its local members (e.g. client C_2); (5) both C_1 and C_2 render the same vote within milliseconds (fig. 5).

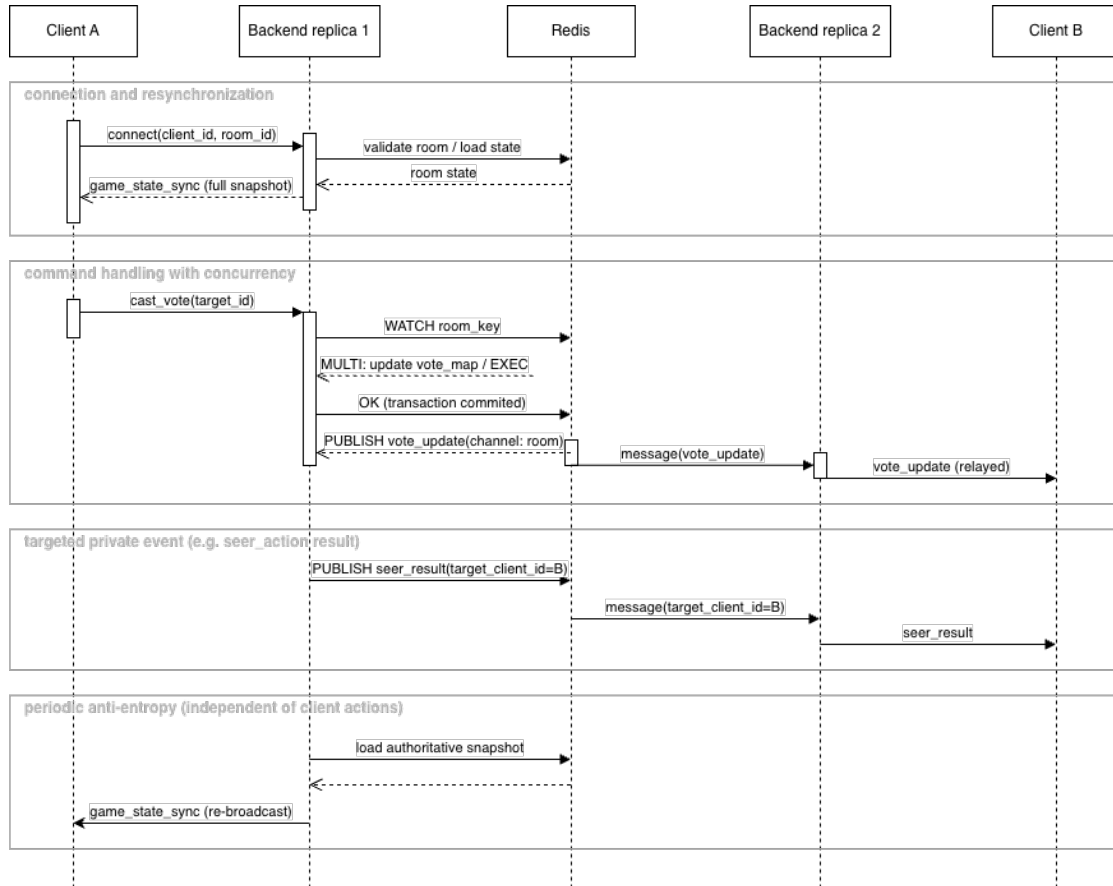


Figure 5: Sequence diagram of a cross-replica game action. The originating replica writes the shared store atomically, emits to its own clients immediately, and publishes on the room channel; another replica relays the event to its local clients, while the originator discards the copy the bus echoes back to it (sender-based deduplication).

Cross-replica unicast handles *private* events (role assignment, seer result), which must reach exactly one client whose location the producer does not know. The event is emitted directly if the target’s socket is local, *and* published with the *target client id* set: each replica receiving it delivers it only if it currently hosts that client’s socket, and never broadcasts it. This preserves both location transparency and secrecy (FR5, FR8).

Periodic anti-entropy broadcast. Because the bus offers at-most-once delivery (section 3.7), a time-driven interaction complements the event flow: every few seconds each replica re-derives the authoritative snapshot of its active rooms and broadcasts it as a *game state sync* event. Clients treat snapshots as absolute truth, overwriting their local view — so any missed event is repaired within one period (NFR3). The same snapshot

is sent point-to-point to every (re)connecting client as the first message (after it supplies its `client_id` and `room_id` in the handshake), making reconnection and initial join the same code path (FR11).

3.5 Behaviour

The match as a replicated state machine. A match evolves through the phase state machine

LOBBY $\rightarrow$ NIGHT $\rightarrow$ DAY $\rightarrow$ VOTING $\rightarrow$ NIGHT $\rightarrow \dots \rightarrow$ ENDED ($\rightarrow$ LOBBY),

where the VOTING $\rightarrow$ NIGHT transition resolves the public lynching, the NIGHT $\rightarrow$ DAY transition resolves the wolf kill and the seer inspection, every resolving transition runs the win check (possibly short-circuiting to ENDED), and the host may loop an ended match back to the lobby. Transitions are triggered by *either* the phase deadline expiring *or* the phase completing early (all living players voted; all wolves and the seer acted) *or* a decisive disconnection.

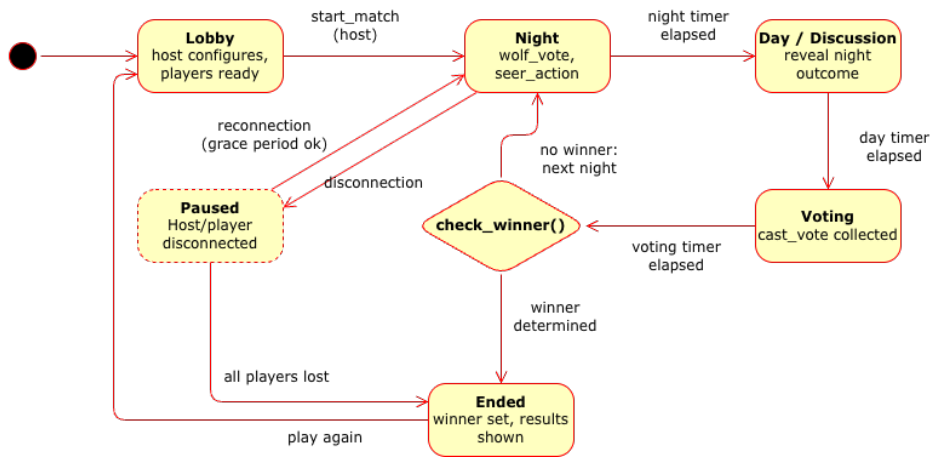


Figure 6: State diagram of a match: the phases and the transitions that resolve the public lynching, the night actions, and the win check, up to the ENDED state and the optional loop back to the lobby.

Backend replicas (stateless, reactive). Each replica is a purely reactive component: it responds to client commands, bus messages, connection lifecycle events, and two internal clocks. Since replicas keep no authoritative state, *any* replica can process *any* trigger for *any* room — correctness is delegated to the store’s atomic operations rather than to replica identity. The two internal clocks are:

- the *phase timers*: when a replica performs a transition it schedules an in-memory timer for the new deadline (the absolute deadline is stored authoritatively). Timers are a local optimisation, not a correctness mechanism: a firing timer checks the *stored* deadline first and aborts if the phase has meanwhile advanced elsewhere (stale-timer guard);

- the *sweeper*: every couple of seconds each replica scans the active-rooms registry for matches whose stored deadline has passed and advances them — so phase progression survives the death of the replica that held the in-memory timer (NFR2).

Both paths (and explicit client-triggered advancement) funnel into a single advancement routine guarded by a short-TTL *distributed lock* acquired atomically in the store: exactly one trigger wins per deadline, so concurrent triggers on different replicas cannot double-advance a phase or eliminate two players (NFR4). The lock is released explicitly on completion through an atomic compare-and-delete (a Lua script that removes the key only if this replica still owns it), so a replica can never release a competitor’s lock. Its short TTL (10s, far below any phase duration) is only a safety fallback should the holder crash mid-advancement.

Clients (stateful views). The client keeps a local replica of the room view and applies events optimistically; on any *game state sync* it discards its view and adopts the snapshot. Its stores are *stateful* (each registers listeners once and updates reactive state per incoming event), while views and visual components are effectively *stateless* (a screen is a pure function of store state); no client component ever writes authoritative game state. On disconnection the client re-connects with the same identity and is treated as a rejoin: mid-game it re-receives its secret role privately and the current snapshot; in the lobby it simply re-enters.

Disconnection: pause, grace, resume. On the server side, a disconnection during a match marks the player *disconnected* but keeps them alive, *pauses* the match, and starts a bounded *grace timer* (20s by default, configurable). The player may reconnect to *any* replica — which cancels the timer (the expiry handler also re-checks the store’s *connected* flag to cover a reconnection landing on a different replica) — and resume their seat, so a transient network drop does not cost the match. Host promotion is *deferred* while the grace timer is pending, so a returning host reclaims its role; only if the timer expires without a reconnection is the player marked as having *fled* — treated as an elimination, to avoid stalling phase-completion checks — after which the match resumes and the win check is re-run (FR9, FR11, FR12). A player who was already eliminated (a spectator) leaving triggers an immediate host promotion and win check without pausing. On the client, this manifests as automatic reconnection, a pause banner while the server signals `game.paused`, and a full resync on `game.resumed`.

Who updates the state. Only backend replicas write to the store, and only in reaction to validated commands or owned transitions; every write is atomic (section 3.6). Clients never write state — they send commands and render events. The proxy and monitoring components are stateless with respect to the domain.

3.6 Data and Consistency Issues

What is stored, and why key–value. All data of section 3.3 is stored in the shared in-memory store as JSON-serialised values under a per-room key namespace: a string key per room snapshot and per game-state record, hashes for the player registry and the ballots (field = player id), a string for the phase deadline, and a set for the active-rooms registry, all with a six-hour TTL renewed on write. A key–value/hash model fits naturally: the data is small, schema-light, always accessed by known key, and never queried relationally — there are no joins, no secondary indexes, no cross-room queries except the lobby list, served by an incremental **SCAN** of the room-snapshot namespace. A relational or document database would add operational weight (IR3) and millisecond-scale latency to a path hit several times per game action. Durability requirements are modest (game state is ephemeral), so an in-memory store with lightweight disk persistence (periodic snapshots plus an append-only log fsynced every second) is an appropriate trade-off: after a store crash, at most the last second of updates is lost — at worst, re-casting a vote.

Who reads and writes. Only backend replicas access the store; the frontend never talks to it directly, which keeps every mutation subject to the authorization checks of section 3.9. Reads happen on every command validation, on every (re)connection (snapshot), on the sweeper/anti-entropy scans, and on lobby-list queries; writes on membership changes, lobby reconfiguration, votes and night actions, and phase transitions. The same room’s state is routinely accessed by several replicas at once (their clients act simultaneously): concurrent reads and writes are the normal case, not an edge case, which is precisely where the consistency issues below arise.

Consistency strategy: strong at the store, eventual at the clients. The design splits the consistency question in two:

- **Authoritative state** must never violate the game invariants (NFR4). All multi-step updates are made *atomic at the store*, which acts as the single serialisation point: every replica’s writes are applied in one total order, so lost updates are prevented by construction rather than by inter-replica coordination. Four mechanisms are used, in increasing strength: (1) *single-operation atomicity* for naturally atomic updates (a vote is one hash-field write); (2) *atomic create-if-absent* (**SET NX**) for room bootstrap — when two strangers create the same room simultaneously on different replicas, only one initial-state write succeeds and its author becomes host, eliminating the “double host” race (FR1); (3) *optimistic transactions* (**WATCH/MULTI**, watch–modify–retry) for read-modify-write patches of the game-state record, aborting and retrying on a concurrent modification instead of silently overwriting it, each successful patch bumping the *state version*; (4) *mutual exclusion* where an entire multi-key procedure must run at most once per deadline — the phase advancement lock of section 3.5.

- **Client views** are only *eventually consistent* [11]: an event can reach clients of different replicas at slightly different times, or be missed entirely (at-most-once bus). The design accepts this — sub-second divergence is imperceptible in a chat-paced game — and guarantees *convergence* instead: the periodic anti-entropy snapshot (section 3.4) overwrites any diverged view within seconds (NFR3), toward the correct value since snapshots are recomputed from authoritative state.

This split is the project’s answer to the CAP trade-off [2]: rather than paying for consensus among replicas [1], the design concentrates consistency in one store and treats replicas as caches holding no divergable copy of domain state at all.

Client-side de-duplication. Because a client can receive the same event through more than one path (its originating replica’s local emit plus a relay, or a re-broadcast after reconnection), the chat store de-duplicates messages by tracking already-seen message identifiers, so a duplicated delivery is rendered only once — a concrete client-side accommodation of the underlying at-least-once delivery that multi-path fan-out produces.

Shared data between components. Beyond domain state, replicas share only two coordination artefacts, both in the store: the active-rooms set (work list for sweeper/anti-entropy) and the advancement locks. Notably, replicas do *not* share their connection registries: which client is connected to which replica is kept local, and cross-replica delivery is solved by publish/subscribe semantics instead of by a shared session directory — one less piece of shared mutable state to keep consistent.

3.7 Fault-Tolerance

The failure model is crash (fail-stop) of any single server-side process, plus message loss: the inter-replica bus provides *at-most-once* delivery, and the network is assumed unreliable in all the classic ways [7].

Replication and redundancy. The *service* is actively replicated (≥ 2 identical replicas, active-active): any replica serves any room, so a crash removes capacity but no function. *Data* is not replicated across nodes (single store); it is made recoverable by the store’s own disk persistence (snapshot + append-only log), restoring state on restart with at most one second of loss. This asymmetry is deliberate: replicating the stateless service is cheap, replicating the store safely is not (primary election and failover), and the dependability requirement — do not lose running matches — is met already if the store’s downtime is short, because all other components mask it.

Heart-beating, timeouts, retries. Failure detection is timeout-based at every seam:

- *orchestrator* $\rightarrow$ *replicas*: liveness probes restart a wedged replica; readiness probes remove a replica from load balancing until it can actually serve (connected to the store), so clients are never routed to a half-alive instance;

- *clients* $\leftrightarrow$ *replicas*: the realtime protocol exchanges application-level heartbeats; a client detecting a dead connection reconnects automatically (through the balancer, hence possibly to a *different* replica) and is re-synced by snapshot;
- *replicas* $\rightarrow$ *store*: every publish is retried a bounded number of times with exponential back-off before being declared lost; the bus *listener* detects connection failure and enters an unbounded reconnect loop with capped back-off, re-subscribing all its channels once the store returns — without this a store blip would silently and permanently cut a replica off the bus;
- *proxy* $\rightarrow$ *replicas*: connect/read timeouts plus request-time DNS re-resolution route around dead replicas.

What happens when each component fails.

A backend replica crashes. Its clients' connections drop; they reconnect through the balancer to surviving replicas and receive the authoritative snapshot — nothing is lost because the replica held nothing authoritative. Its in-memory timers die with it: the cross-replica sweeper notices the expired stored deadline within a couple of seconds and advances the match on another replica, under the advancement lock (NFR2).

The store becomes unreachable. Commands requiring state fail cleanly (clients get error events and can retry); already-delivered views remain rendered, so ongoing discussion is not interrupted. Each replica's listener keeps retrying; on recovery, publishes resume, channels are re-subscribed, and the next anti-entropy round re-syncs every client. Events lost on the bus during the outage are repaired by the same mechanism — the design never assumes bus reliability, only store reliability *over time*.

The proxy or frontend crashes. Both are stateless; the orchestrator restarts them, and clients experience a transient connection error. The monitoring stack is entirely out of the game's fault path.

Table 3: Failure modes and how the platform tolerates them. Detection is timeout-based at every seam; recovery never depends on the failed component itself, and the fault of any single server-side component leaves running matches intact.

Failure	Detected by	Recovery	Player impact
Backend replica crash	orchestrator liveness/readiness probes; client heartbeat	clients reconnect through the balancer to a surviving replica and re-sync by snapshot; the sweeper advances the phase timer under the advancement lock	brief reconnect; match continues (NFR2)
Store (Redis) unreachable	Pub/Sub listener error; failed store commands	listener reconnect loop with back-off + channel re-subscription; publish retries; anti-entropy re-syncs every client on recovery	state-changing commands rejected briefly; rendered views persist
Proxy / frontend crash	orchestrator liveness probe	stateless restart; request-time DNS re-resolution routes around it	transient connection error, auto-reconnect
Client disconnect / network drop	socket close + missed heartbeats	match paused, grace timer started; reconnect (to any replica) resumes it, otherwise the player is dropped	short pause; resumes on return
Monitoring stack down	Prometheus self-monitoring / alerts	stateless restart	none (observability only)

Error handling within the protocol. Invalid or unauthorised commands (wrong phase, non-host configuration attempt, dead voter) are rejected by validation against authoritative state and answered with a private *error* event — misbehaving or buggy clients cannot corrupt state. Unexpected exceptions in event handlers are caught per-event: one malformed message cannot take down a replica’s event loop. Corrupted records read from the store (failed deserialisation) are logged and skipped rather than crashing the reader.

3.8 Availability

Caching. There is no cache tier by design: the store is itself in-memory with sub-millisecond access, so a cache in front of it would add a consistency problem without a latency benefit. Two cache-like mechanisms exist nonetheless: the per-room *snapshot* record is a denormalised, ready-to-send materialised view (refreshed on every relevant change and by anti-entropy), so client synchronisation is one read; and each *client* caches the room view locally between events, keeping the UI responsive during a brief server-side outage. Static SPA assets are served with standard HTTP caching.

Load balancing. Load balancing happens at connection granularity at the entry proxy: new realtime connections are assigned to backend replicas *round-robin*. Round-robin is right for long-lived connections: each connection is a lasting resource commitment, so the goal is to spread *connection counts* evenly; latency-aware policies were rejected because a persistent connection produces no response-time signal until it closes. Because clients reconnect through the balancer, load *re*-balances naturally after failures and scale-ups. Horizontal elasticity complements balancing: replicas scale automatically between a floor of 2 (fault tolerance) and a ceiling of 8, scaling up aggressively on load spikes and down slowly — draining a replica disconnects its clients, so scale-down is damped to once a minute after a 5-minute stabilisation window (NFR5, NFR8). The ceiling of 8 is a pragmatic cap rather than a hard limit: past a point the real bottleneck becomes the single Redis instance shared by every replica as both state store and bus, so adding backends only shifts pressure onto it. Eight replicas — at roughly 80 concurrent WebSocket connections per replica, some 640 simultaneous players — comfortably exceed the load a casual party game must sustain while staying within the single-machine, zero-budget target (minikube, IR3). Lifting the ceiling meaningfully would first require a highly available store (section 9).

Behaviour under network partition. Applying CAP [2] per partition scenario, the system fails soft rather than closed:

Client partitioned from the cluster. The match goes on: phases advance server-side regardless of individual clients (timers are authoritative), and the partitioned player's inaction is handled by the same rules as any inaction; on prolonged disconnection the player is marked as having fled. On heal the client re-syncs by snapshot — rebuilding from the snapshot, never from any locally cached delta. This favours availability for the *group* over the individual, the correct choice for a party game.

Replica partitioned from the store. The replica keeps its clients connected but refuses state-changing commands (it cannot validate them): it degrades to read-only availability on stale views rather than guessing — for authoritative writes the design chooses consistency over availability, because a wrong elimination is worse than a delayed one. Its readiness probe fails, so the balancer stops routing *new* connections to it. Phase timers are protected by comparing the locally scheduled deadline against the stored deadline before acting, preventing a previously-partitioned replica from advancing a phase twice.

Split between replicas (bus unreachable, store reachable). Impossible to sustain silently in this topology (bus and store are the same component), but the at-most-once analysis of section 3.7 covers transient versions: views diverge for at most one anti-entropy period, then converge. Writes remain correct throughout, since they serialise at the store.

Independently of any partition scenario, the Kubernetes deployment is configured for backend high availability: two replicas, rolling updates without downtime, and readi-

ness/liveness/startup probes together with a `preStop` hook that drains in-flight connections before a pod is terminated (NFR2, NFR8).

3.9 Security

Authentication. There is no account system: players self-identify with a nickname scoped to a room (a `client_id` generated and stored client-side, passed in the Socket.IO handshake), which doubles as their client identifier. This is a conscious trade-off (section 2.5): the product is a casual party game, no sensitive data is at stake, and login friction would hurt the primary use case more than impersonation would. The nickname is weakly protected where it matters for gameplay: a name already connected to a room cannot be taken by a second connection (preventing casual impersonation and accidental double tabs, FR14), and the same identity lets a disconnected player resume their seat (FR11).

Authorization. Access control is enforced *server-side on every command*, on authoritative state, never on client claims. It is a small role-based scheme with three axes: *platform role* (host-only rights — configure, start, kick, restart — checked against the stored host identity); *game role and liveness* (only living players may vote; only wolves may wolf-vote; only the seer may inspect; spectators retain read access only); and *phase* (every command is valid only in its phase — configuration in LOBBY, votes in VOTING, night actions in NIGHT), which blocks whole classes of out-of-order exploits. On the client, chat channels are filtered by role/liveness and action controls are enabled only for the eligible role in the correct phase — but this client-side filtering is a UX safeguard only; the authoritative enforcement is server-side. Information-flow control complements access control: secret data (roles, wolf ballots, seer results) is delivered only over private unicast events and is systematically excluded from broadcast snapshots until game end — so even a modified client observing all its traffic learns nothing it should not (FR5, FR8).

Perimeter and cryptography. The proxy is the only exposed component; internal components are on segregated networks and unreachable from outside (section 3.2). It applies per-IP rate limits (separate budgets for API calls and connection attempts) to blunt abuse and trivial denial-of-service, sets standard security headers, and hides implementation details. No custom cryptographic schema is used — there are no passwords or tokens to protect, and transport encryption (TLS) is a deployment concern the proxy is already prepared for (port 443 is reserved; the local course deployments run plain HTTP). Interactive API documentation is disabled outside development.

4 Implementation

This section binds the design of section 3 to concrete technologies, motivating each choice.

4.1 Network protocols

Two application-level protocols are used, one per interaction kind of section 3.4. Client–server realtime communication uses **Socket.IO** [9] on top of WebSocket (over TCP/HTTP). Socket.IO was preferred to raw WebSocket because it provides out of the box exactly the machinery the design requires: named events, *rooms* (the per-room topic abstraction), application-level heartbeats (ping/pong) for failure detection, and automatic client reconnection with back-off — re-implementing these reliably over raw WebSocket is error-prone and out of scope. The HTTP long-polling fallback that Socket.IO offers is deliberately *disabled* on the client (WebSocket-only transport), to keep a single, clean upgrade path behind the reverse proxy. Queries (lobby list, health) and the single initial-state fetch use plain **HTTP/1.1** REST endpoints.

Between backend replicas, synchronization reuses neither Socket.IO nor HTTP: it happens over **Redis Pub/Sub**, i.e. the Redis serialization protocol (RESP) over TCP, the same connection already used for state persistence [6]. Reusing the existing Redis connection for both state and cross-replica notification avoids introducing a separate message broker, which would add an infrastructural component and a failure mode without a corresponding benefit at this scale. The backend runs as a single ASGI process exposing both the REST and the Socket.IO surface; the proxy upgrades WebSocket connections (**Upgrade/Connection** headers) and keeps them alive with hour-long read timeouts.

4.2 In-transit data representation

All payloads — client events, bus messages, stored values — are **JSON**. The dominant consumer is a browser, so JSON is the zero-friction choice, and Socket.IO is built around JSON-friendly payloads on the JavaScript side. Payloads are small (well under the enforced 1 MB cap) and exchanged at most a handful of times per second per room, so a binary format (Protocol Buffers, MessagePack) would buy little except schema tooling on both the Python backend and the Vue frontend. Using the same representation for the wire format and for the Redis-persisted state also lets a payload be stored and broadcast without an intermediate transformation. Message *schemas* are nonetheless explicit in code: every event family has a typed model — Pydantic models for the **RedisEvent** bus envelope (event id, type, room, sender replica, timestamp, payload, optional target client) and the **WSMessage** client envelope, plus dataclasses per event payload converted to JSON-serializable dictionaries before emission — so malformed messages are rejected at the boundary rather than deep in game logic.

4.3 Database querying

The store is **Redis 7** [6], accessed through its native asynchronous Python client (**redis.asyncio**) — no ORM and no SQL, since the data model is key–value by design (section 3.6). All interaction is NoSQL: key–value and hash operations (**GET**, **SETEX**, **SET NX EX**, **HGET**, **HGETALL**, **HSET**, **SADD**, **SMEMBERS**), combined with **WATCH/MULTI/EXEC** transactions for read-modify-write sequences and a Lua script for atomically releasing the phase-advancement

lock. The consistency mechanisms of section 3.6 map to Redis primitives one-to-one: atomic creation is `SET NX EX`; optimistic transactions are `WATCH/MULTI/EXEC` retried on `WatchError`; the advancement lock is `SET NX` with a 10s TTL released by the compare-and-delete Lua script; bulk player updates use pipelines (one round-trip instead of $O(n)$); the lobby list uses cursor-based `SCAN` rather than the blocking `KEYS`. TTLs (`SETEX/EXPIRE`, 6h, renewed on write) delegate garbage collection of abandoned rooms to Redis instead of a scheduled task, keeping replicas as stateless as possible. Pub/Sub channels are named `game:{room_id}` plus one global channel.

4.4 Authentication

As designed in section 3.9, there is no OAuth/JWT machinery. The client generates (or retrieves, if already present) a `client_id`, stores it in `sessionStorage`, and sends it — together with the room id — in the Socket.IO connection handshake; the back-end treats this identifier as a session identity rather than a cryptographic credential, binding it server-side to the socket id via the connection manager. This is a deliberate simplification, adequate for the controlled, ephemeral environment the game runs in; it would need to be replaced by a real authentication mechanism before the system could be exposed where impersonating another player's `client_id` carried real consequence.

4.5 Authorization

Authorization is implemented as guard clauses at the top of every command handler, reading the authoritative Redis state (host id, phase, role, liveness) and raising domain errors that reach the offending client as private *error* events. At the lobby level, only the host may update settings or start the match. During a match, gameplay actions are enabled only in the phase they belong to and only for entitled players: `cast_vote` is accepted only during `VOTING`, while `wolf_vote` and `seer_action` are accepted only during `NIGHT` and only from players holding the corresponding role, with additional checks that the player is still alive and has not already acted in the current phase. An RBAC framework would be over-engineering for three roles and a dozen rules.

4.6 Technological details

Backend: Python 3.12, FastAPI, python-socketio, Uvicorn. The backend is an ASGI application composed of **FastAPI** [5] (REST endpoints, dependency-injected settings via `pydantic-settings`, OpenAPI docs in development) and **python-socketio**'s `AsyncServer` mounted on the same server, run by **Uvicorn** with the `uvloop` event loop. An asynchronous single-threaded runtime is the natural match for the workload (section 2.5): thousands of mostly idle sockets multiplexed over one event loop, with I/O-bound handlers awaiting Redis. Each container runs *one* Uvicorn worker — scaling is horizontal at the container level (section 6), which keeps per-process state (connection registry, timers) trivially consistent and lets the orchestrator own scaling. Long-running concerns (Pub/Sub listener, timer sweeper, anti-entropy) are `asyncio` background tasks started in FastAPI's lifespan hook. Each replica identifies itself with an `INSTANCE_ID` taken

from the `HOSTNAME` environment variable (the pod/container name), implementing the sender-id deduplication and lock ownership of section 3 with zero configuration.

Frontend: Vue 3, Pinia, Vue Router, Vite, socket.io-client. The SPA uses **Vue 3** (Composition API, `<script setup>`), built and served in development by **Vite**, with **Vue Router** mapping four routes to the four screens (home, lobby, game, results) and **Pinia** for state management. The three stores of section 3.3 (lobby, game, chat) each expose reactive state, computed selectors (alive players, own role, whether an action is currently allowed, the visible chat messages), the event listeners, and the command-emitting actions. Real-time communication goes through a single **socket.io-client** connection shared application-wide (one module-level instance): WebSocket-only transport; an authentication payload carrying the `client_id` and room id; automatic reconnection with a bounded number of attempts; and a pending-listener registry that lets stores register handlers before the socket exists and re-binds them when the connection opens, avoiding mount/connection races. Listeners are de-registered/re-registered around reconnections which, combined with message de-duplication (section 3.6), prevents duplicated or lost chat messages. The UI is composed of small reusable components: a player card (SVG with a hidden back or a revealed role), a deterministic avatar (colour and initials from the player id), a chat box, a circular phase timer whose colour changes as the deadline approaches, a role-icon, and an info box. The backend origin is not baked in at build time: a `config.js` is generated *at container start* by an entrypoint script (`envsubst` on a template, reading the `WS_URL` environment variable, defaulting to same-origin), so the same image runs under `localhost`, `game.local`, or any production domain (build-once/configure-at-runtime).

Containerisation. Both images are multi-stage builds: the backend separates a dependency-compilation stage from a slim runtime (`python:3.12-slim`); the frontend builds with `node:20-alpine` and ships only compiled assets on `nginx:1.25-alpine`. Both run as non-root users and embed container-level health checks, yielding small, reproducible, least-privilege images — the unit of deployment everywhere from Compose to Kubernetes.

Edge and observability stack. The reverse proxy is **NGINX 1.25** (Compose) and the **NGINX Ingress Controller** (Kubernetes), configured as per sections 3.2 and 3.8. Metrics use `prometheus-fastapi-instrumentator` for HTTP metrics plus custom counters/histograms (table 4), all labelled by `instance_id`; **Prometheus** [4] scrapes replicas and a `redis_exporter`, evaluates recording/alert rules (backend down, Redis down, error rates), and **Grafana** ships with a provisioned dashboard — so observability (NFR7) works out of the box in both deployments.

Table 4: Custom application metrics exported on `/metrics` (standard HTTP metrics are added automatically by the instrumentator). Every metric additionally carries an `instance_id` label, so all quantities can be broken down per replica; the *Labels* column lists only the further dimensions.

Metric	Type	Labels	Measures
<i>WebSocket connections</i>			
<code>ws_active_connections</code>	Gauge	—	live WS connections on the replica
<code>ws_connections_total</code>	Counter	—	connections accepted since start
<code>ws_disconnections_total</code>	Counter	<code>reason</code>	disconnections (normal / timeout / error)
<i>WebSocket messages</i>			
<code>ws_messages_received_total</code>	Counter	<code>event_type</code>	client → server messages, by event
<code>ws_messages_sent_total</code>	Counter	<code>event_type</code>	server → client messages, by event
<code>ws_message_size_bytes</code>	Histogram	—	size of received messages (bytes)
<i>Redis Pub/Sub bus</i>			
<code>redis_messages_published_total</code>	Counter	<code>channel</code>	events published on the bus
<code>redis_messages_received_total</code>	Counter	<code>channel</code>	events received from the bus
<code>redis_messages_deduplicated_total</code>	Counter	—	own-echo events dropped (dedup)
<code>redis_publish_failures_total</code>	Counter	—	publishes lost after all retries
<code>redis_reconnects_total</code>	Counter	—	listener reconnects to the store
<i>Rooms & players</i>			
<code>game_active_rooms</code>	Gauge	—	rooms with ≥ 1 local client
<code>game_active_players</code>	Gauge	—	connected players on the replica
<i>Internal latency</i>			
<code>redis_publish_duration_seconds</code>	Histogram	—	PUBLISH latency to the broker
<code>ws_broadcast_duration_seconds</code>	Histogram	—	time to broadcast to a room

5 Validation

Validation combines three layers of automated testing — unit, integration, and end-to-end — with manual acceptance testing of the parts that automated tests do not cover well. The layers go from pure logic to the deployed system, and deployment automation

(section 6) makes it cheap to bring up a production-like environment to test against.

5.1 Automatic Testing

5.1.1 Unit testing

Domain logic and the Redis access layer are unit-tested in isolation, using `fakeredis` — an in-process emulator honouring the same command semantics — in place of a real Redis instance, so tests exercise the *real* data-access code (including transactions) without any running infrastructure: fast, deterministic, and CI-friendly. On the back-end, `test_game_logic.py` covers role assignment and its validation bounds (FR3, FR5), phase transitions and their resolutions — vote tallying with ties and skips, night kills, seer inspection, win detection (FR6–FR9); `test_lobby_logic.py` covers host election, ready-state handling, and lobby rules (FR12); `test_state_store.py` covers persistence and retrieval through the Redis access layer. The frontend mirrors this with **Vitest**: store tests verify how the three Pinia stores react to incoming events (a `phase_changed` updates the phase, a `vote_update` updates the tally, chat de-duplication discards an already-seen message — mapping onto FR2–FR8), plus composable/router tests for the countdown and the route configuration. Views are intentionally not unit-tested: a screen is a pure projection of store state (already tested) and is better validated end-to-end. Corner cases live here — a player with no assigned role yet, an already-expired phase timer, a match at the minimum player count, an empty initial lobby.

5.1.2 Integration testing

A second layer verifies interaction between modules that cooperate on the same state within a single replica, and the Redis concurrency primitives of section 3.6 under the conditions they were designed for: `test_lobby_runtime.py` and `test_backend_init.py` check that the runtime emits the expected events for a given command sequence, while `test_distributed_safety.py` targets the distributed guarantees directly, one guarantee per test — atomic room creation electing exactly one host under a simulated double-join (FR1, NFR4); the advancement lock admitting a single winner per room while remaining independent across rooms (NFR4); optimistic patches merging concurrent field updates and bumping the `state_version` monotonically, i.e. no lost updates (NFR4); the active-rooms registry behaving as the sweeper expects (NFR2); and bulk pipeline persistence preserving per-player data. These properties are exactly the ones manual testing cannot reliably exercise: the tests encode the race conditions as deterministic scenarios.

5.1.3 End-to-end / system testing

Standalone Python scripts open genuine Socket.IO/WebSocket connections against a running backend (Compose with two backend replicas, or the Kubernetes cluster), exercising the system as a whole rather than through direct function calls, using the same `socket.io` client as the real frontend:

- `test_multi_instance.py` validates replication transparency (NFR1, FR10): events published by a client on replica *A* reach clients on replica *B* and vice versa, local broadcast still works, and the sender never receives its own event twice (deduplication). It can target the load-balanced entry point or two specific pods via `kubectl port-forward`, asserting on the `instance_id` carried by every message.
- `test_reconnection.py` validates state synchronisation (FR11, NFR3): a fresh room yields no spurious snapshot; a client reconnecting to the same or a *different* replica receives the state produced while it was away; two clients on different replicas observe consistent state after churn.
- `load_test.py` validates NFR5/NFR6: a multi-process Socket.IO load generator (hundreds of concurrent clients spread over several OS processes, so that the bottleneck under test is the platform, not the generator) measures connection success and event delivery under load, and is used to observe HPA scale-up and connection spreading live (`kubectl get hpa -w` and the Grafana dashboard). A companion diagnostic (`diag_loss.py`) distinguishes real event loss from test artefacts.

The corner cases exercised here include loss or delay of Pub/Sub messages, a client reconnecting to a different replica, a duplicate-host race, both local and cross-instance broadcast, bursts of messages in a short window, and disconnection followed by state recovery. Crash-recovery of phase timers is exercised by killing a backend pod during a scripted match (the sweeper advances the phase within its period — NFR2). Lightweight smoke checks confirm reachability before the more expensive runs:

```
# Smoke checks
curl http://game.local/health
wscat -c "ws://game.local/ws/test_room?client_id=test123"

# Multi-instance propagation (replication transparency)
python tests/test_multi_instance.py --url http://game.local
# Reconnection and state consistency
python tests/test_reconnection.py --url http://game.local
# Message-loss diagnostics
python tests/diag_loss.py ws://game.local/ws
# Load test
python tests/load_test.py --url http://game.local --clients 300 --procs 4
```

Layer	What it validates	Rationale	Requirement(s)
Unit	Domain rules, state serialization, UI derivations, in isolation via <code>fakeredis</code> / mocks	Fast, deterministic feedback on pure logic and corner cases	Underlies NFR1; FR3–FR9, FR12
Integration	Cooperating runtime modules and Redis concurrency primitives on a single replica	Individually-correct components can still misbehave once combined at sensitive points	FR1, NFR2, NFR4
End-to-end	Real network communication, cross-replica propagation, reconnection, load	Only observable with a running system, not from function calls	NFR1–NFR6, FR10–FR11

Table 5: Summary of the automated testing layers.

5.2 Acceptance test

Manual play-testing sessions were run on the full deployment for the properties that resist automation, because they involve several real clients interacting over the network in real time. Independent client identities were obtained with multiple browser windows in separate incognito sessions. The scenarios exercised:

- *full-game flows*: complete matches with 5+ players including seer play, skip votes, mid-game abandonments, kicks, end-game screens, and play-again loops (FR4–FR9, FR13) — automating a whole match with hidden-role coordination across many scripted clients was judged poor value, since the underlying rules are already unit-tested;
- *perceived experience*: UI readability, timer smoothness, and the simultaneity of phase changes and eliminations across screens — qualitative judgements (layout, role banners, perceived responsiveness) a script cannot make;
- *fault and multi-instance scenarios*: transient disconnection and reconnection during a match (verifying state resynchronisation), players of the same room deliberately spread across two backend replicas (confirming events reach everyone and the client behaves identically regardless of routing), and operational drills — killing pods mid-match, restarting Redis, and watching the system converge (semi-automated via deployment scripts + dashboards, validated by human observation).

Each acceptance session mapped observed behaviour back to the acceptance criteria of section 2; all functional requirements were verified on the Kubernetes deployment before delivery.

6 Deployment

The platform supports three deployment profiles, all driven by files versioned in the repository; complete step-by-step instructions live in the repository `README`. Table 6 summarises them. They share the same images and configuration mechanism (section 4.6) and differ only in what is containerised and in the target environment.

Table 6: The three deployment profiles.

Profile	Bring-up command	Brings up	Entry point
1 — Development <i>support only, hot-reload</i>	<code>docker compose -f docker-compose-dev.yml up -d + native uvicorn --reload and npm run dev</code>	Redis, exporter, Prometheus, Grafana (backend and frontend run natively)	Vite dev server on <code>localhost</code>
2 — Full Compose <i>single machine, full stack</i>	<code>docker compose up -d --scale backend=2</code>	NGINX, <i>N</i> backend replicas, frontend, Redis, monitoring stack	<code>http://localhost</code>
3 — Kubernetes <i>cluster, production-like</i>	<code>k8s/deploy.sh</code> (or the documented <code>kubectl apply</code> sequence)	namespace, ConfigMap/Secret, Redis + PVC, backend + HPA (2–8), frontend, monitoring, Ingress	<code>http://game.local</code> (via <code>minikube tunnel</code>)

6.1 Prerequisites and configuration

The only host requirements are Docker Desktop 24+ with Compose v2 and, for the cluster profile, `kubectl` 1.28+ and `minikube` 1.32+ (with the `ingress` and `metrics-server` addons). `Node.js` 20 and `Python` 3.12 are needed only to run or test frontend/backend outside their containers. These versions match the base images actually used: the backend is built from `python:3.12-slim`, the frontend from `node:20-alpine`, and the infrastructure containers use `redis:7.2-alpine`, `prometheus:2.51.0`, and `grafana:10.4.0`.

All application configuration flows through *environment variables* with sensible defaults (Redis URL, channel prefix, phase durations, log level), already defined — with working values — directly inside the deployment manifests (`docker-compose.yml`, `docker-compose-dev.yml` and the `k8s/` ConfigMaps). No variable needs to be set by hand and no separate `.env` file must be created for a standard installation: running `docker compose up` or `kubectl apply -f k8s/` is enough on its own. The same images run in every environment (section 4.6).

6.2 Profile 1 — development (support services only)

`docker compose -f docker-compose-dev.yml up -d` starts only the stateful/support services (Redis, its exporter, Prometheus, Grafana), while the backend (`uvicorn main:app`

--reload) and frontend (npm run dev) run natively with hot-reload:

```
docker compose -f docker-compose-dev.yml up -d
cd backend && pip install -r requirements.txt && uvicorn main:app --reload --port 8000
cd frontend && npm install && npm run dev
```

This keeps the edit cycle at IDE speed while developing against real infrastructure; it trades the production-like guarantees of the full stack (reverse proxy, multiple replicas, containerised frontend) for fast reloads and is not the final deployment target.

6.3 Profile 2 — full Docker Compose

docker compose up -d --scale backend=2 builds and starts the entire platform on one machine: NGINX (the only published entry point, port 80), N backend replicas, the frontend, Redis, and the monitoring stack. The backend is explicitly scaled to more than one replica so that the distributed behaviour of section 3 is actually exercised rather than degenerating to a single instance:

```
docker compose build
docker compose up -d --scale backend=2
docker compose ps
curl http://localhost/health
```

The Compose file encodes the design decisions of section 3.2 directly: four segregated bridge networks (proxy–frontend, proxy–backend, backend–Redis, monitoring); named volumes for Redis, Prometheus, and Grafana; health checks on every service with `depends_on: condition: service_healthy` (backends start only after Redis answers PING); a replica-friendly backend definition (no fixed container name or published port, so `--scale backend= $N$` just works, NGINX resolving the service name at request time); and resource limits mirroring the Kubernetes ones. Expected outcome: every service *healthy*; the game at `http://localhost`, the WebSocket endpoint at `ws://localhost/ws/<room_id>`, the REST API under `http://localhost/api/`, Prometheus at `:9090`, Grafana at `:3001`; `curl /health` returns each replica’s id in round-robin.

6.4 Profile 3 — Kubernetes (minikube)

Images are built into minikube’s Docker daemon, then either the `k8s/deploy.sh` script (or its PowerShell twin) or the documented `kubectl apply` sequence deploys, in dependency order — namespace, ConfigMap/Secret, Redis (waiting for rollout), backend, frontend, monitoring, Ingress:

```
kubectl apply -f k8s/namespace.yml
kubectl apply -f k8s/configmap.yml
kubectl apply -f k8s/redis/
kubectl apply -f k8s/backend/
```

```
kubectl apply -f k8s/frontend/
kubectl apply -f k8s/monitoring/redis-exporter.yml
kubectl apply -f k8s/monitoring/prometheus.yml
kubectl apply -f k8s/monitoring/grafana.yml
kubectl apply -f k8s/ingress.yml
# or, equivalently:
./k8s/deploy.sh
```

The manifests add production-grade concerns on top of the Compose design (table 7): a backend Deployment with zero-downtime rolling updates and three probes; an HPA scaling 2–8 replicas; a single Redis with a PersistentVolumeClaim and **Recreate** strategy (never two writers on one volume); a WebSocket-ready Ingress with per-IP limits, plus a *separate* monitoring Ingress without those limits so dashboards stay reachable during load tests; and Prometheus rules and the Grafana dashboard injected as ConfigMaps generated from the same source files used by Compose (a single source of truth). Expected outcome: `kubectl get pods -n game` shows `backend` $\times 2$, `frontend` $\times 2$, `redis`, `redis-exporter`, `prometheus`, `grafana` all **Running**; with `minikube tunnel` active and `game.local` in the hosts file, the game answers at `http://game.local`, `/health` returns `ok`, and Grafana/Prometheus are reachable under `/grafana` and `/prometheus`. Verify with:

```
kubectl get pods -n game
kubectl get services -n game
kubectl get ingress -n game
kubectl get hpa -n game
curl http://game.local/health
```


Table 7: Key configuration of the main Kubernetes objects (namespace `game`). Values are taken directly from the manifests; the rationale column links each choice to the requirement it serves.

Object	Key configuration	Rationale
Backend Deployment	2 replicas; rolling update <code>maxSurge=1</code> , <code>maxUnavailable=0</code> ; <code>terminationGracePeriod</code> 60 s + <code>preStop</code> 5 s drain; requests 250m/256Mi, limits 1 core/512Mi; three probes (startup 30 × 5 s, liveness <code>/health/20</code> s, readiness <code>/health/10</code> s)	zero-downtime updates (NFR8); no traffic to a replica not yet connected to Redis (NFR2)
HorizontalPodAutoscaler	min 2 / max 8; CPU 50% + memory 70%; scale-up +2 pods/15 s (no window); scale-down −1 pod/60 s (300 s window); <code>ws_active_connections</code> metric prepared but disabled	elastic capacity, damped to protect WS connections (NFR5)
Redis Deployment + PVC	1 replica; <code>Recreate</code> strategy; 1 Gi <code>ReadWriteOnce</code> PVC; requests 100m/128Mi, limits 500m/512Mi; RDB + AOF <code>everysec</code> , <code>maxmemory</code> 256 mb <code>volatile-lru</code>	single stateful store; never two writers on the volume; disk persistence for recovery
Ingress (game)	<code>round_robin</code> balancing; proxy read/send timeout 3600 s; buffering off; body size 1 m; per-IP <code>limit-connections</code> 1000, <code>limit-rps</code> 300	WebSocket-ready edge with perimeter rate limiting (NFR6, security)
Ingress (monitoring)	separate object, without the per-IP limits	dashboards stay reachable during load tests

7 User Guide

7.1 Application workflow overview

The application follows a linear, intentionally simple workflow that guides the user through four consecutive macro-phases, **Home** → **Lobby** → **Game** → **Results**, directly reflected in the frontend routing. The game UI is in Italian; the walkthrough below names the on-screen labels where useful.

7.2 Usage

Starting the application. Point a browser at the entry URL: `http://localhost` with Compose, `http://game.local` on Kubernetes. The home page offers two paths: *Crea Partita* (create a new game) and *Entra in Lobby* (join an existing lobby).

Creating a new lobby. Selecting *Crea Partita* and entering a nickname creates a new room in which the user automatically becomes the host, and produces a shareable room code.

Joining an existing lobby. Selecting *Entra in Lobby*, the user enters a nickname (3–15 characters) and either types a room code or picks one from the live list of open lobbies (with host and player count), then confirms.

Preparing the game. Inside the lobby the room code is shown with copy-to-clipboard (to invite friends). Only the host can configure the match (number of wolves and optional seers) or kick players; everyone else toggles a *ready* flag. When at least 5 players are connected and all are ready, the host’s start button unlocks and the match begins — each player privately receives their role card. If the host leaves, another player is promoted automatically.

Playing the game. The game screen shows the population (player cards with liveness), the current phase with its countdown, the chat, the player’s own role, and the actions allowed by phase and role:

- **Night:** villagers sleep (“Stai dormendo”); wolves see their pack, vote a victim, and coordinate in a dedicated pack chat (“Chat del Branco”); the seer picks a player to inspect (“Sfera di Cristallo”) and privately learns their faction.
- **Day:** the night’s outcome is announced and everyone discusses in chat.
- **Voting:** every living player votes a suspect or skips; votes are visible as they arrive. The most-voted player is eliminated and their role revealed; ties and skip majorities spare everyone.

A phase ends when its timer expires or as soon as every awaited action has been performed. Eliminated players remain as spectators.

Handling disconnections and reconnections. The system transparently handles temporary interruptions: on a page refresh or a lost connection the client automatically restores the session using the `client_id` stored in `sessionStorage`. During a match, losing the connection (closing or reloading the tab, a network drop) *pauses* the match and opens a short grace window: reconnecting within it — even onto a different server replica — restores the session automatically and the match resumes; only if the window elapses without a return is the player treated as having fled and eliminated, so the game can continue for everyone else.

Viewing the final results. When a faction wins, the results screen (“Identità rivelate”) announces the winning team, reveals every player’s role, and shows the match summary (rounds played, survivors, eliminated, wolves). The host can then bring the whole group back to the lobby with one click to play again with the same room; other players can return to the lobby or home.

7.3 Expected outcomes

Home. Two clearly distinguishable call-to-action buttons (*Crea Partita*, *Entra in Lobby*), the live list of available lobbies when joining, and validation of the entered nickname before proceeding.

Lobby. A clearly visible room code, the connected players with each one's ready status, the role-configuration panel, and a start button enabled only for the host and only when all conditions hold; the lobby updates in real time over WebSocket.

Game. Real-time updates of the current phase, a timer synchronised across all clients, actions varying by assigned role, an integrated chat, and clear visual feedback for votes, night actions, and reconnections; actions not allowed by the current role or phase are rejected by the backend.

Results. A clear indication of the winning team, the revelation of every player's role, a summary of the match statistics, and the possibility to start another match (host) or return to lobby/home (other players).

7.4 Visual walkthrough

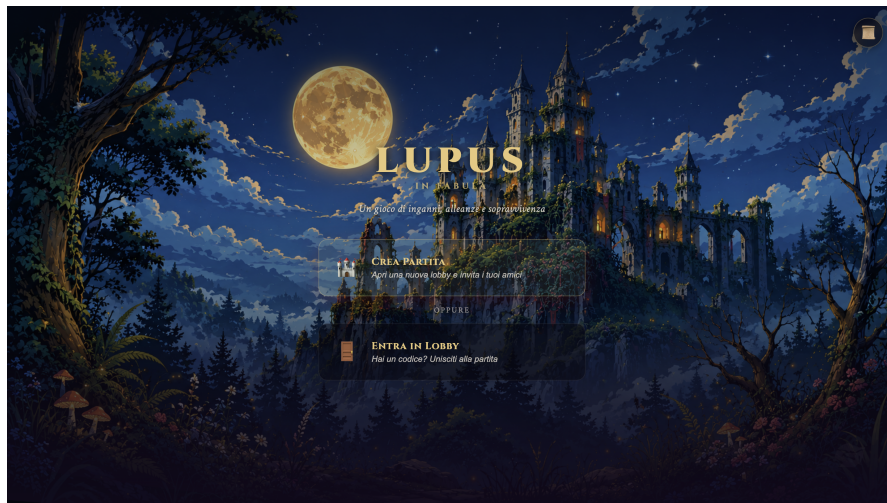


Figure 7: Home page: the two main options, *Crea Partita* and *Entra in Lobby*.

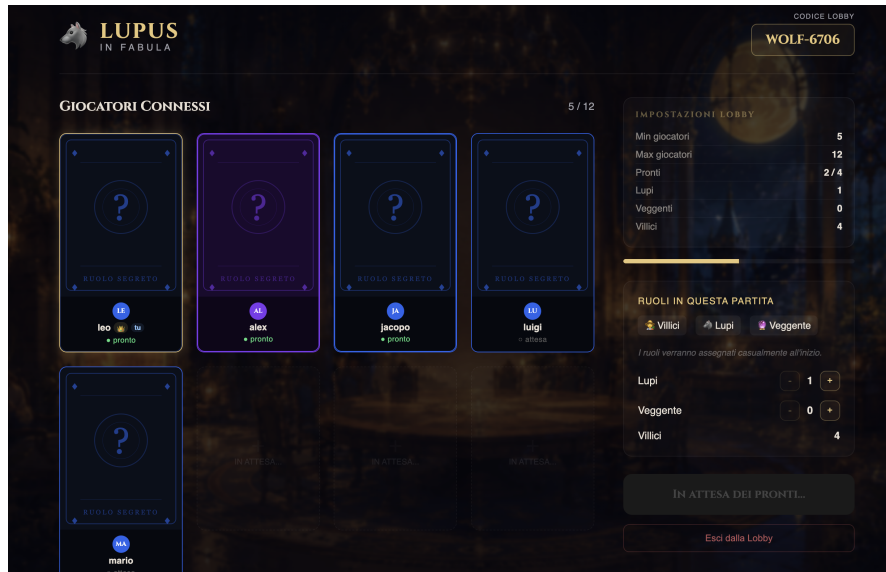


Figure 8: Lobby screen: room code, connected players and their *Ready* status, and the game-configuration panel.

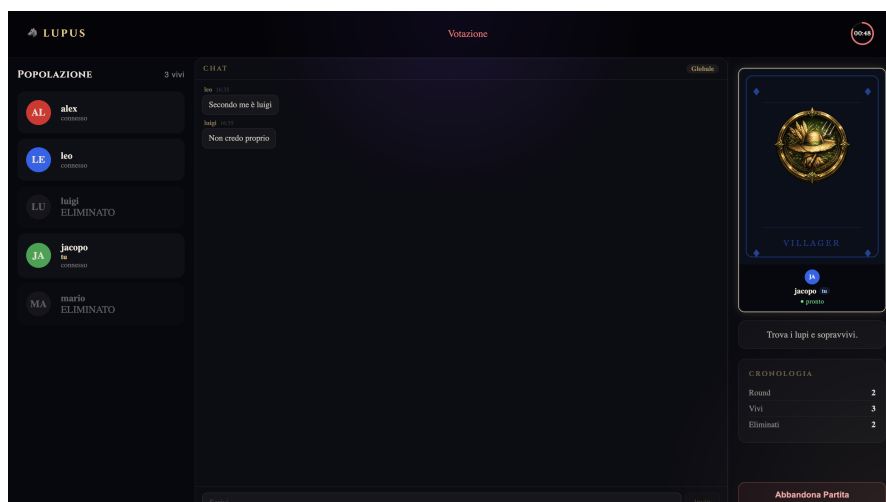


Figure 9: Game screen during the day phase, while voting is in progress.

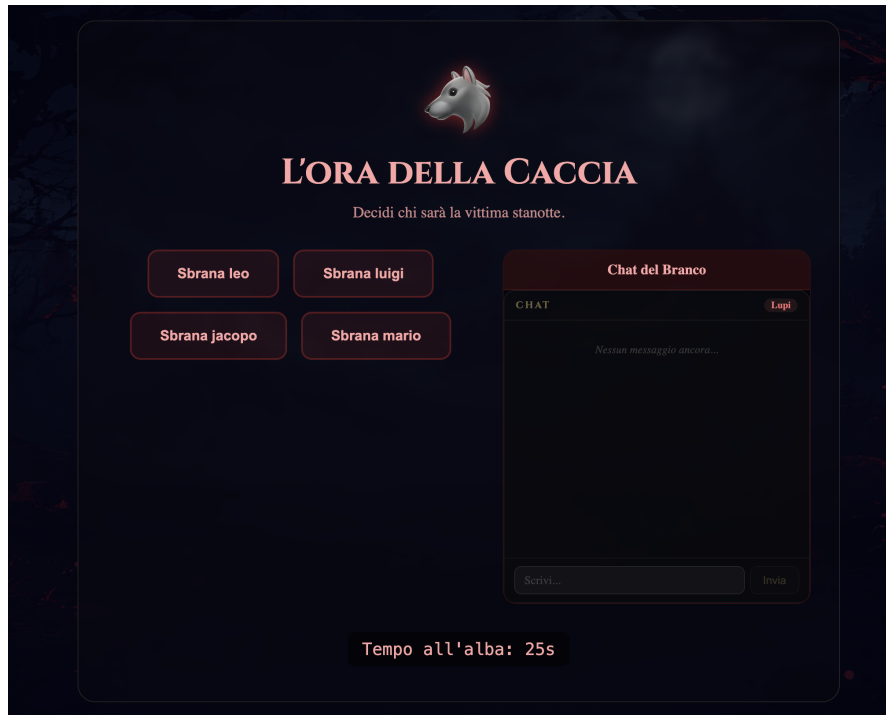


Figure 10: Game screen during the night phase, showing the actions available only to the wolves and the seer.



Figure 11: Final results screen announcing the winning team and revealing all player roles.

8 Self-evaluation

Each member is responsible only for their own subsection.

8.1 Jacopo Bonifazi — Frontend & Lobby

My role in the group was to design and implement the entire frontend and the lobby system: the user interface (home, lobby, game, and results screens), the client-side state management with three Pinia stores, the integration with the real-time channel via the Socket.IO client, and the client-side handling of disconnection and recovery. I also owned the definition of the Socket.IO event contract on the client side, in coordination with the backend.

Strengths of the product. The client is cleanly layered and event-driven, which keeps game logic entirely on the server and makes the UI a pure projection of the authoritative state; this made the resynchronisation after a fault straightforward and robust. The restricted-visibility chat and the per-role, per-phase action gating meet the functional requirements, and the message de-duplication correctly handles the at-least-once delivery that arises in the multi-instance deployment.

Weaknesses / limitations. The pause/resume experience depends on server-side events that were still being finalised, so its full behaviour was not completely exercised. The component-level UI is validated mostly by manual end-to-end testing rather than automated component tests. Some auxiliary REST scaffolding remained unused once the interaction consolidated on the event channel.

Contribution. Objectively, I delivered the full frontend and lobby as planned, and additionally helped diagnose several integration issues that surfaced in the containerised, multi-instance environment (e.g. the WebSocket endpoint mismatch and the instance-local player counting), which, although resolved server-side/infrastructure-side, were identified from the client behaviour.

8.2 Alexandru Nicolescu — Infrastructure & Distribution

My role in the group was the *distribution and infrastructure* side of the project: I designed and built the deployment layer (the Docker images and Compose topology with its segregated networks and health checks, the NGINX reverse-proxy configuration for WebSocket balancing, the full set of Kubernetes manifests with HPA and rolling updates, and the Prometheus/Grafana monitoring stack), I led the migration of the realtime layer from raw WebSocket to Socket.IO, and I implemented the cross-replica machinery: the Redis Pub/Sub manager with sender-based deduplication, retries and reconnection, and the distributed-safety mechanisms (atomic host election, phase-advancement lock,

optimistic state patches, timer sweeper, anti-entropy). I also wrote the infrastructure-level validation: the multi-instance, reconnection, and distributed-safety suites, and the multi-process load generator.

The main strength of this part of the product, in my view, is that replication is real and tested, not nominal: the platform genuinely survives replica kills mid-match, and every claimed guarantee has a test that encodes its race condition. I am also satisfied with the symmetry between the Compose and Kubernetes environments, which made every production-like experiment cheap.

The main weaknesses I recognise are, first, that Redis remains a single stateful point of failure — mitigated by persistence and by how the rest of the system degrades, but not eliminated (an HA store was beyond the time budget, see section 9); and second, that some convergence guarantees rest on periodic anti-entropy rather than on acknowledged delivery, which is adequate for a game but required care to reason about and would not transfer as-is to a domain with stricter requirements. Working through those trade-offs — and discovering how many “obvious” single-node behaviours (timers, session counting, private messages) silently break under replication — was the most instructive part of the course project for me.

8.3 Leonardo Vorabbi — Backend

Within the group, I was responsible for the backend of the project, focusing on the server-side logic of the application. My main responsibilities included designing and implementing the backend architecture, managing game and lobby state, implementing the Redis-based persistence layer, handling Socket.IO events and real-time server–client communication, managing phase transitions and timers, defining the game logic and validation rules for actions, voting, and role-based behavior, and implementing and maintaining the backend test suite.

From my point of view, the main strengths of the product lie in its stateless-at-process-level backend architecture, which makes horizontal scaling practical, and in the hybrid communication model, combining REST for simple queries with Socket.IO/WebSocket for real-time gameplay. The system supports multiple backend replicas and keeps working correctly even when players are connected to different instances. The codebase is organized into clearly separated layers (models, state management, game logic, transport), and is backed by a test suite covering unit tests, integration tests, distributed-safety checks, and end-to-end scenarios.

The main weaknesses concern authentication, which is intentionally lightweight and based on client-side session identifiers, and could therefore be unsuitable for a production environment requiring strong identity management.

9 Future works

The current implementation already provides a complete distributed real-time game platform, but several directions remain open to strengthen robustness, simplify maintenance, and extend gameplay.

High-availability store. The single Redis instance is the one component whose failure pauses the whole platform (section 3.7). The natural evolution is Redis Sentinel (primary/replica with automatic failover) — the client library already supports it, so the change is confined to configuration and manifests — or, at larger scale, Redis Cluster with per-room key hashtags so that each room’s keys stay on one shard and the existing transactions keep working.

Strong authentication and persistent identities. Client identity is currently a lightweight value in `sessionStorage` passed in the Socket.IO handshake. A production-grade solution would introduce a JWT/OAuth login flow, require the signed token on both REST and Socket.IO connections, and persist user profiles (username, avatar, match history) in a database, making host privileges and player identity independent of client-provided values.

At-least-once event delivery. Convergence currently relies on periodic anti-entropy over an at-most-once bus. Replacing room channels with *Redis Streams* and consumer groups would give persistent, acknowledged, replayable per-room event logs — turning reconnection into a replay from the last seen event id and making the snapshot re-broadcast a rare repair path instead of a periodic one.

Connection-aware autoscaling. The HPA currently scales on CPU/memory, which underrepresents a connection-bound workload (section 3.8). The backend already exports an active-connections metric per replica, and the manifest documents the Pods-metric block to scale on it; enabling it only requires installing the Prometheus Adapter in the cluster.

Persistence of chat and match history. Chat messages currently live only in the real-time channel, with no long-term history. Storing them in a database or a per-room Redis list, adding pagination, and persisting match summaries (phases, eliminations, winners) would enable a history/replay page and improve debuggability.

Richer matchmaking and lobby management. The lobby workflow could be extended with public/private rooms and filters, automatic room assignment, invite links or QR codes, and room templates for different rule sets.

More roles and configurable rule variants. The fixed core role set and phase structure could gain roles such as Hunter, Doctor, or Cupid, with role composition and game mode externalised into a configuration file selectable by the host before the match. The event-based design keeps this cheap: a new role is a new payload type, new validation rules, and new private events, with no change to the distribution machinery.

Frontend polish. Complete the pause/resume UX once the corresponding server events are finalised (including a graceful “player did not return” outcome), add automated component/end-to-end tests to cover the UI flows currently checked manually, remove the unused REST scaffolding now that interaction is event-based, and add richer, always-legible error feedback on the entry screen.

Security hardening and operational polish. TLS termination at the proxy/Ingress (ports and headers are already prepared), Redis authentication (the Secret template exists), stricter CORS and WebSocket origin checks, rate limiting on sensitive actions, and protected monitoring endpoints; plus a CI pipeline running the three test rings on every push, image publication to a registry with digest pinning, and structured JSON logging shipped to a log aggregator to complement metrics.

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